

Real Time Facial Emotion Recognition using Deep Learning and CNN

S.K. LokeshNaik *

Department of Computer Science and Engineering
MLR Institute of Technology
Hyderabad, India
sklokeshnaik@gmail.com

Ajmeera Kiran

Department of Computer Science and Engineering
MLR Institute of Technology
Hyderabad, India
sklokeshnaik@gmail.com

M.Sudheeksha

UG Student
Department of Computer Science and Engineering
MLR Institute of Technology
Hyderabad, India
minenisudheeksha14@gmail.com

A.Punitha,

Associate Professor
Department of Computer Science and Engineering
Annamalai University, TamilNadu, India
12charuka17@gmail.com

A. Narsimha Dhangar

UG Student
Department of Computer Science and Engineering
MLR Institute of Technology
Hyderabad, India
narsimhachintu20@gmail.com

P.Vijayakarthish

Professor,
Dept of Computer Science and Engineering,
R.L.Jalappa Institute of Technology,
Dodabalapur, Bengaluru.
vijayakarthish@rljit.in
B.Jathin Reddy
UG Student
Department of Computer Science and Engineering
MLR Institute of Technology
Hyderabad, India
reddyjathin95@gmail.com

Abstract— In this universe, a person's mood may be read from their facial expressions. But one thing is for sure: every expression is a way that people communicate their emotions. Despite the fact that every person has a unique set of facial features and a range of emotions, artificial intelligence (AI) technology allows us to recognize these emotions. A face emotion-recognition AI is built using a variety of techniques, one of which is deep learning to build a convolutional neural network (CNN) to train the data set and obtain high prediction accuracy. Recognizing seven of the fundamental facial emotions out of hundreds is the major goal here. The most common facial emotions are those of happiness, sadness, neutrality, anger, surprise, fear, and disgust. There are plenty of ways to achieve this facial emotion recognition like using deep face algorithms, machine learning, by AI techniques, by Tensor flow etc. The accuracy of our model is between 70% to 80%. It is perhaps the simplest and efficient approach of all of them.

Keywords- Deep Learning (DL), Convolutional Neural Network (CNN), Face Detection, Haar Cascade, and Emotion Detection

I. INTRODUCTION

People's feelings can be instantly conveyed through their facial expressions in a moment's notice. Research in the fields of computer vision and artificial intelligence have found that procedures that recognise emotions based on facial expressions have an important role [1]. Even though emotion identification can be done using wearable sensors, doing it with visual inputs rather than having a physical link is preferable because it is both more important and more versatile [2]. The rising use of artificial intelligence systems can be attributed to the developments that have occurred in computer technology in recent years. Realizing emotion identification based on facial expressions is highly crucial in real-time applications where humans and computers interact, such as humanoid robot applications that use artificial intelligence systems. One example of this type of application is video games. As a result of advancements in face emotion detection technology, applications in a variety of fields, including the medical field, the retail industry, and

the entertainment industry, have made significant headway [3].

Facial expression is the universal language that all humans use to communicate their emotions. A person's manner of expression might reveal a great deal about the way they are feeling at the moment. The intention is to convey the feeling that a person is experiencing through the use of live video streaming. The dataset contains over 20,000 photographs of human faces, each of which has been labelled on a scale ranging from 0 to 6. The labels 0, 1, 2, 3 represent angry, disgusted, fearful, and pleased emotion, while the labels 4, 5, 6 suggest sad, surprised, and neutral emotions respectively. Our Convolutional Neural Network model takes this enormous training dataset as an input, and then trains the dataset itself based on the labels. In contrast, the Viola-Jones Face Detection Technique, more commonly referred to as Haar Cascades, is a classifier that is utilised for the purpose of face detection. Despite the fact that there are a number of Haar cascade classifiers for detecting eyes, mouths, and noses. However, for this particular technology, the only face classifier that we have used is the Haar cascade frontal face classifier. We are making use of a real-time video stream from a webcam in order to display our predicted emotions on the face of the subject. To do this, we are superimposing a square over the subject's face while simultaneously displaying the emotion on the screen.

Because the majority of the datasets used in studies involving face emotion identification do not contain a sufficient number of photos, facial emotion detection is a challenging application to implement. The vast majority of research on emotion identification in real time has been conducted on still photographs. When looking at the research published over the past two decades, there has been a discernible increase in the number of studies that use computer vision techniques to attempt to recognise facial expressions of emotion. This rise has occurred in direct correlation with the advancement of computer technology. Convolutional neural networks, also known as CNNs, are not typically favoured for the visualisation of visual properties in the majority of applications relating to face recognition and detection [4]. This is despite the fact that there are many applications that deal with face recognition and detection. Visual methods were utilised by Breuer and Kimmel [5] in order to gain an understanding of a CNN model that was trained with a range of datasets for the purpose of emotion recognition. They test the performance of CNN using facial emotion recognition datasets as well as several applications that use facial emotion recognition. A method that makes use of two distinct varieties of CNNs was devised by Jung and colleagues [6]. Extraction of features from visual datasets is one of them, while extraction of the geometric properties of facial landmarks is the other. Both of these processes are carried out in parallel. An example of one of the uncommon experiments on video was presented by Kahou et al. [7], who trained deep convolutional neural networks by first applying them to

visual datasets and then applying them to video. In this article, we propose an emotion detection system that uses deep learning and is based on the analysis of facial expressions taken from video frames in real time. In contrast to applications of a similar nature, the system maintains a high level of accuracy even while working with a variety of models and in a variety of environments. An original dataset was developed for six fundamental feelings, including anger, fear, joyful, neutral, sad, and surprise respectively. The CNN model was then trained using this dataset as the input. Experimental research utilising this dataset were also carried out on AlexNet, which is a pre-trained network. In addition to that, the CNNs that were obtained were evaluated against a standard set of test data.

II. RELATED WORKS

The expression of one's feelings is one of the most fundamental ways in which humans communicate with one another. It is a crucial element of the nonverbal communication that we engage in on a daily basis. [1] The technique of determining human emotions, most typically through facial expressions, is referred to as "emotion recognition," and the name "emotion recognition" refers to this process. In recent years, a range of different approaches have been put to the test in an effort to classify human emotions based on images of faces [2, 3], [4]. In addition to this, a sizeable portion of them have achieved decent levels of accuracy [5, 6, 7]. A wide variety of expression recognition systems and other systems that are able to recognise the movement of muscles on an individual's face are able to classify a person's emotions, including happiness, sadness, anger, and a variety of other emotions. These systems are able to do this because they are able to recognise the movement of the muscles on an individual's face. Characterizing the movements of the facial muscles is something that the Face Action Coding System (FACS), which is widely regarded as one of the most useful psychological frameworks, does exceptionally well [8]. The topologies of GoogleNet and AlexNet were the inspiration for the work that was presented in [9]. Their network is composed of two distinct parts that are linked together in some way. The first network consists of two traditional CNN modules connected together. After this, there are two more modules designed in the same fashion as Inception. These are carried out in parallel and include a 1x1, 3x3, and 5x5 convolution layer that all make use of the ReLU algorithm. After that, each of these layers is brought together to create the output by being connected together. They utilise two FC layers as their classification layers, and both of these FC layers utilise ReLU in some capacity. [10] explains how the model being proposed is more like a network, complete with seven distinct layers. It consists of one full-connected layer, one softmax layer, two layered pooling layers with the labels P1 and P2, three convolutional layers, one full-connected layer, and three softmax layers. As the network's source of data, we will provide it with a face intensity pixel matrix that is 48 by 48 in size. Both the convolutional layer and the pooling layer each have their own collection of unique feature maps within their respective layers. The feature map of each succeeding layer

is connected to the feature map of the layer below it in a fashion that is only partially connected. Convolution layers C1, C2, and C3 each have their own unique set of 64, 64, and 128 convolution kernels to use when performing their respective operations. Convolution kernels with a size of 5x5 are utilised for both C1 and C2, whereas convolution kernels with a size of 4x4 are utilised for C3. The overall window size corresponds to the sum of the sizes of the windows on the pooling levels P1 and P2, which are each 2 by 2. It is said that the fully connected layer is "fully linked" because it has 4608 neurons and is completely connected to the pooling layer P2 below it. Seven neurons make up the softmax layer, and they are all accountable for the feature classification that occurs in the fully connected layer. The study that is referred to in [11] utilised an architecture that consists of two convolutional layers, followed by maxpool, a convolutional layer, dropout, maxpool, three fully connected layers (each followed by a dropout), and finally a softmax layer. This architecture was used to train a machine learning model. The 19-layer VGGNet network model that has been modified is the one that is used in [12]. Within the context of this specific model, the convolutional layer can be broken down into a total of five distinct parts. Each segment is comprised of two to four convolutional layers, and the final layer of each segment is a maximum pooling layer. This layer is connected to the following segment in order to assist in the process of decreasing the size of the entire image. In the first segment, there are 64 convolution kernels, in the second segment there are 256, in the third segment there are 512, and in the fourth segment there are also 512. This indicates that there are exactly the same number of convolution kernels present in each section. The very first layer with a full connection is the only one that is maintained; subsequent levels have dropout layers placed into the empty areas between the layers with full connections. After the fully-connected layer comes the eventual addition of a softmax classifier as a subsequent link in the model's chain. The cross-entropy loss function and the Softmax activation function are both incorporated into this classifier. The authors of "13" have created CNNs with varied levels of depth for their research. During the first phase of the learning process, a shallow CNN consisting of one convolution layer and three fully-connected layers is utilised. Rectified Linear Units, also known as ReLUs, are used as the activation function in each and every one of the layers. This is done across the board. Utilizing a deep convolutional neural network (CNN) is the next step in the process. A deep CNN has a total of 9 layers: 3 fully connected levels, 4 max pooling layers, and 8 convolutional layers. The first and second convolutional layers each consist of a total of 32 3x3 filters, making up the total number 32. When it comes to the third and fourth convolutional layers, a total of 64 filters measuring 3x3 are put to use. The fifth and sixth convolutional layers each have 96 filters of a size of 3x3, for a total of 192 filters. The seventh and eighth convolutional layers each have 128 filters that are 3 by 3, and they are composed of these filters. The authors of [14] generate a CNN that is specific to their data by following the preprocessing procedure that they have developed. The very last part of the model consists of a

collection of layers, all of which are connected to one another. As a basis for its construction, it makes use of the Xception architecture. The development of this model makes use of both residual modules and deep-separable convolutions as building blocks in the process. It is a fully convolutional neural network with four deep-separable residual convolutions present in each layer. This is followed by batch normalisation and ReLU activation, which completes the network. In the very last layer, a softmax activation function is used, and a global average is computed before being used in the training process. A DCNN model is used in the analysis presented in [15]. The structure of VGG16 served as an inspiration for the design of this model. Convolution, Max-Pooling, and Fully Connected Layer Operations Were Iterated Numerous Times The final output of seven classes was accomplished by repeatedly applying Convolution, Max-Pooling, and Fully Connected Layer Operations. Every one of the pooling layers has a pool size of 2x2, while the convolution layers all have a filter size of 3x3 for their layers. An activation layer comes immediately after each and every one of the convolution layers. After each activation layer, the batch normalisation process is carried out so that the end output would have a higher degree of accuracy.

III. EXISTING ARCHITECHTURE

For the purpose of this investigation, the steps listed below are carried out. A dataset containing six fundamental feelings was compiled using internet photos that are open to the public and may be found using various search engines. The Viola-Jones method is applied to these photos as part of the face identification process in order to discover face positions on the images and crop them in preparation for the creation of new image files. The dataset is broken up into two distinct sections: the training dataset, and the test dataset. The training dataset is used to teach the CNN, while the test dataset is used to evaluate the network's effectiveness after it has been trained. In the end, the resultant network is put to use in order to categorise the facial expressions depicted in the video frames. Figure 1 and 2 illustrates the study's flowchart of existing and proposed , which may be found below.

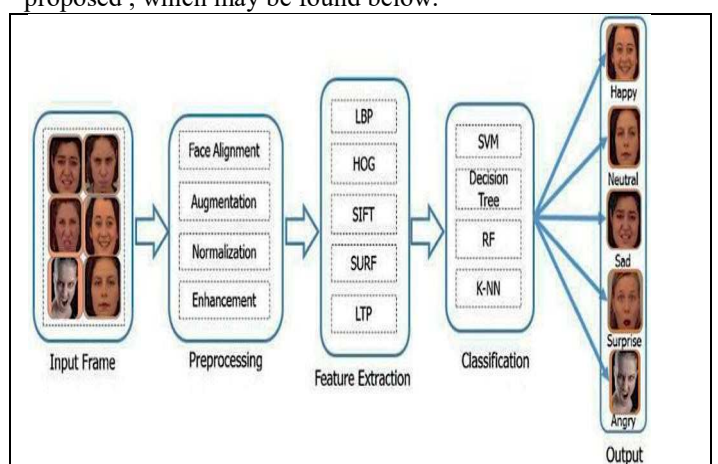


Fig.1.ExistingArchitecture

Datasets and Functionalities: The dataset that is being utilised in this investigation was constructed from publicly available photos [8] that were made available without charge. These pictures have been labelled with common face expressions such as anger, fear, pleased, neutral, sad, and surprise. After removing some of the images that were incorrectly labelled, a total of 600 images were obtained with a smallest margin of 240 pixels for each class. These photographs are used to compile a dataset that consists of 3600 images that are saved in the RGB format. Following that step, this dataset is then split into a training dataset and a test dataset. The selection process resulted in 3360 photos being used for the training dataset and 240 images being used for the test dataset.

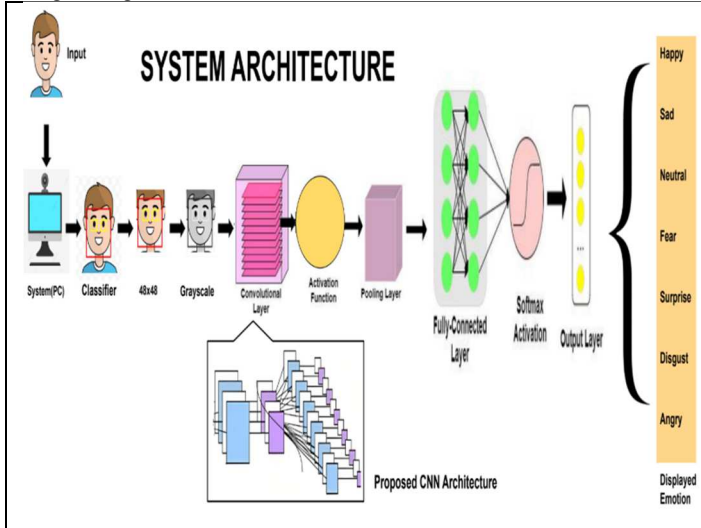


Fig.2.ProposedArchitecture

IV. RESULTS AND DISCUSSION

OpenCV has been used to develop a face detector that is both quick and reliable for use in face detection (which was initially used by P. Viola and M. Jones). When presented with a real-time live stream, the face detector investigates each area of the image to determine whether or not it contains a face in that particular place. The data contained in an XML file is read by the classifier and used to make decisions regarding how to categorise each image location. OpenCV's Haar Cascade classifier file is selected as the one to use for XML data detection. A number of frames are taken from the camera, and then face detection is performed on each of those frames. The bounding box list is ordered according to the size of the bounding box, with the faces that have the largest bounding boxes appearing at the beginning of the list. This is provided that at least one face has been identified. The region of interest (ROI) is determined by selecting the area of the face that is the largest, and then the image is grayscale and resized to be 48 by 48 pixels. The expression on the face is analysed in order to determine which label corresponds to it with the highest related probability. The live stream is displayed in a window that has a bounding box around the largest face, and the emotion label is displayed directly above the bounding box in real time.

V. CONCLUSION

The construction of a deep neural network capable of recognising facial expressions was the objective of this body of work. We have gone over the steps necessary to set up a Convolutional Neural Network that is able to analyse and predict facial expressions and emotions. In order to complete this mission, a CNN that works similarly to VGG has been trained and evaluated. There is a good chance that improved accuracy can be achieved by employing a wider variety of data augmentation techniques, elevating the network's depth, expanding the number of its layers, and including additional regularisation. The purpose of this research is to provide an artificial (deep network) convolution network model for the recognition of face emotions. The image dataset is used to train the machine to recognise seven distinct face expressions. Both the training accuracy and the validation accuracy of the model are comparable, which indicates that the model is having the best fit possible and is able to generalise to the data. The accuracy of the model has been tested to be between 70 and 80 percent, and it makes use of an Adam optimizer to cut down on the loss function. This article has covered a variety of topics, ranging from emotions and how they are associated with facial expressions to how computers read faces to the various methods that can be used to successfully execute facial emotion detection.

REFERENCES

- [1] Duncan, Dan, Gautam Shine, and Chris English. "Facial emotion recognition in real time." Stanford University (2016).
- [2] J. D. Bodapati and N. Veeranjanyulu, "Facial Emotion Recognition Using Deep Cnn Based Features", 2019 International Journal of Innovative Technology and Exploring Engineering (IJITEE), Volume-8, Issue-7, pp. 1928-1931.
- [3] S. Miao, H. Xu, Z. Han and Y. Zhu, "Recognizing Facial Expressions Using a Shallow Convolutional Neural Network," in IEEE Access, vol. 7, pp. 78000-78011, 2019.
- [4] Minaee, S., and A. Abdolrashidi. "Deep-emotion: Facial expression recognition using attentional convolutional network. arXiv 2019." arXiv preprint arXiv: 1902.01019.
- [5] Revina, I.M., Emmanuel, W.R.S. A Survey on Human Face Expression Recognition Techniques. Journal of King Saud University Computer and Information Sciences (2018)
- [6] Hashim Abdulsalam, Wisal & Salam, Mohammed & Al-hamdani, Dr. (2019). Facial Emotion Recognition from Videos Using Deep Convolutional Neural Networks. 9. 14-19.
- [7] Chuanhe Liu, Tianhao Tang, Kui Lv, and Minghao Wang. 2018. MultiFeature Based Emotion Recognition for Video Clips. In Proceedings of the 20th ACM International Conference on Multimodal Interaction (ICMI '18). ACM, New York, NY, USA, 630-634.
- [8] B. C. Ko, A Brief Review of Facial Emotion Recognition Based on Visual Information. Sensors (2018), 18, 401.
- [9] A. Mollahosseini, D. Chan and M. H. Mahoor, "Going deeper in facial expression recognition using deep neural networks," 2016 IEEE Winter Conference on Applications of Computer Vision (WACV), Lake Placid, NY, 2016, pp. 1-10.
- [10] L. Xu, M. Fei, W. Zhou and A. Yang, "Face Expression Recognition Based on Convolutional Neural Network*," 2018 Australian & New Zealand Control Conference (ANZCC), Melbourne, VIC, 2018, pp. 115-118.
- [11] Minh-An Quinn, Grant Sivesind, Guilherme Reis, "Real-time Emotion Recognition From Facial Expressions", 2017, CS229 Stanford University.
- [12] H. Jun, L. Shuai, S. Jinming, L. Yue, W. Jingwei and J. Peng, "Facial Expression Recognition Based on VGGNet Convolutional Neural Network," 2018 Chinese Automation Congress (CAC), Xi'an, China, 2018, pp. 4146-4151.
- [13] R. Pathar, A. Adivarekar, A. Mishra and A. Deshmukh, "Human Emotion Recognition using Convolutional Neural Network in Real Time,"

2019 1st International Conference on Innovations in Information and Communication Technology (ICIICT), CHENNAI, India, 2019, pp. 1-7.

[14] A. M. Ousmane, T. Djara, F. J. Zoumarou W. and A. Vianou, "Automatic recognition system of emotions expressed through the face using machine learning: Application to police interrogation simulation," 2019 3rd International Conference on Bio-engineering for Smart Technologies (BioSMART), Paris, France, 2019, pp. 1-4.

[15] M. I. U. Haque and D. Valles, "A Facial Expression Recognition Approach Using DCNN for Autistic Children to Identify Emotions," 2018 IEEE 9th Annual Information Technology, Electronics and Mobile Communication Conference (IEMCON), Vancouver, BC, 2018, pp. 546-551.